914528

Course Name: COMPUTER NETWORKS

Course Outcome

CO1- Understand the concept of OSI and TCP/IP reference model.
CO2- Understand the basics of data transmission at physical layer.
CO3- Understand the channel allocation using ALOHA, CSMA and CSMA/CD
CO4- Apply error detection and correction technique to eliminate transmission error.
CO5- Analyze the fixed and variable length address (IPv4) subnetting for the given scenarios.
CO6- Understand the design issues of the transport layer.
CO7- Understand the mechanism of protocols at application layer such as FTP, HTTP, Telnet, DNS.
CO8- Understand IPv6 addressing and differentiate it from IPv4.

Printed Pages :2 **University Roll No. .....**

Mid Term Examination, Odd Semester 2025-26
Program (CSE), 2nd Year, III Semester
BCSC0008 & COMPUTER NETWORKS

Time: 2 Hours **Maximum Marks: 30**

Instruction for students:

1. All parts of the sections should be answered at same place.
2. Answer should be brief and to the point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Two computers are communicating over the internet. The sender application generates a message, which is encrypted before transmission. During transmission, the message is broken into packets, routed across multiple networks, and finally reassembled at the destination. Identify which OSI layers are primarily responsible for: (a) Encryption (b) Packet Routing (c) Message Reassembly. Suppose A sender transmits data at 10 Mbps, but the receiver can only process at 2 Mbps. If there is no flow control mechanism, what problem will occur?	3	1	U	C
2	A network operates using slotted ALOHA with a channel capacity of 100 kbps. If frames are of size 200 bits each, calculate the maximum throughput and explain why slotted ALOHA performs better than pure ALOHA.	3	3	An	A
3	Differentiate between the following transmission media in terms of bandwidth, attenuation, and cost: a) Twisted pair	3	2	U	C

	b) Co-axial cable c) Optical fiber				
4	A link has a transmission speed of 10^6 bits/second. It uses data packets of size 1000 bytes each. Assume that the acknowledgment has negligible transmission delay and that its propagation delay is the same as the data propagation delay. Also assume that the processing delays at nodes are negligible. If the efficiency of the stop-and-wait protocol in this setup is exactly 25%, then what is the value of the one-way propagation delay (in milliseconds).	3	3	An	A
5	A small software company is setting up a wired Local Area Network (LAN) with 25 desktop systems connected through a shared Ethernet cable. The network administrator is considering different medium access methods. a. Explain how Carrier Sense Multiple Access (CSMA) and Carrier Sense Multiple Access with Collision Detection (CSMA/CD) would work in this setup. b. The company is also planning to add a Wi-Fi network for mobile devices. Why is CSMA/CD not suitable for wireless communication? c. Compare the performance of CSMA and ALOHA by giving an example scenario (e.g., when 10 employees attempt to send data simultaneously). Which protocol is expected to perform better, and why?	3	3	An	A

Section – B

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Compare the Go-Back-N ARQ and Selective Repeat ARQ protocols. a. If the sender has a window size of 4 and frames 1, 2, 3 are sent successfully but frame 4 is lost, explain what happens in each protocol.	5	4	An	A

	b. Calculate the maximum window size for both protocols if the sequence number field is m = 4 bits.				
7	A sender needs to send the four data items 0x3456, 0xABCC, 0x02BC, and 0xEEEE. Answer the following: a. Find the checksum at the sender site. b. Find the checksum at the receiver site if there is no error. c. Find the checksum at the receiver site if the second data item is changed to 0xABCE.	5	4	A	P
8	A data stream 1101001 is to be transmitted using three different line coding schemes: 1. NRZ-L (Non-Return-to-Zero Level) 2. Manchester Coding 3. Differential Manchester Coding (a) Draw the waveform for the given sequence in all three coding schemes. (b) Which scheme is self-synchronizing and why? (c) If the data rate is 1 Mbps, what is the minimum bandwidth requirement for Manchester coding?	5	3	An	C